GROWTH AND DEVELOPMENT

YOUNG CHILDREN DEVELOP BASIC PHYSICAL SKILLS AND THEN BECOME MORE AGILE, WITH INCREASED INTELLECTUAL ABILITIES. PHYSICAL GROWTH RATE IS RAPID DURING INFANCY, AND THEN IS FAIRLY STEADY UNTIL IT SPEEDS UP AGAIN AT PUBERTY.

BONE GROWTH

Body growth depends on the increasing size of the skeleton. The long leg bones provide most of the increase in height. Many long bones develop from cartilage precursors, by a sequence of changes (ossification) that starts before birth at primary centres in the bone shafts. After birth, secondary centres develop near the bone ends. Growth ceases once ossification is complete, at 18–20 years of age.

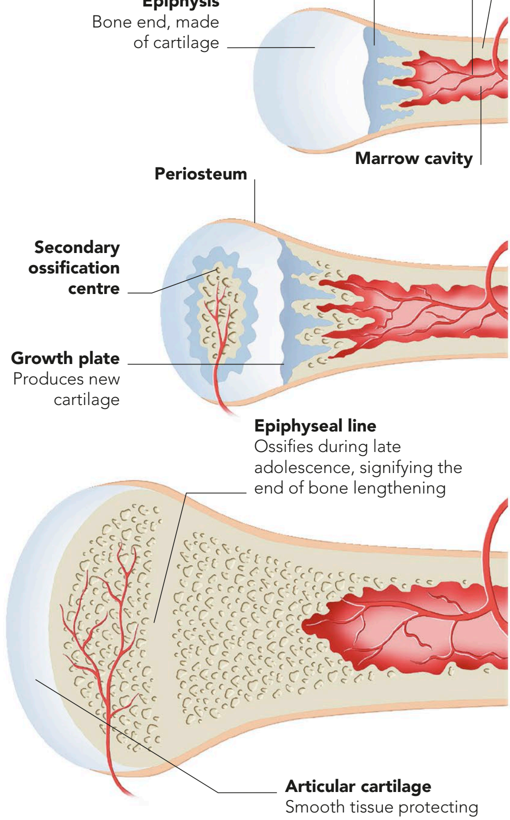

1 YEAR


3 YEARS


20 YEARS

CARTILAGE TO BONE

These X-rays of the hand and wrist show bony tissue as darker pink, blue, or purple. At one year, the bones in the wrist and fingers are largely made of cartilage. At three years, ossification of cartilage is well under way, and by 20 years the bones are fully formed.



LONG BONE OF A NEWBORN

The shaft turns to hard bone from the primary ossification centre, and has a marrow cavity. The head is all cartilage, and is relatively soft.

LONG BONE OF A CHILD

A secondary ossification centre inside the head begins to change the surrounding cartilage to hardened, mineralized bone. An elongating growth area (growth plate) forms between the shaft and head.

LONG BONE OF AN ADULT

By about 18–20 years, all zones have hardened into true bone, with the epiphyseal growth plate represented by a line of dense bony tissue. The only remaining cartilage is smooth and slippery articular cartilage, which covers the head of the bone inside the joint.